

Module 4

Chapter 4

Expectation

What is the idea behind Expectation

For *numerical random variables*, we may be interested in a *single value* that captures what to expect from the variable – the *average result* if it's sampled many times. For example, the expected value, in dollars, of a lottery is how much we should expect to gain or lose on average every time we play. If we're talking about an algorithm with a random component, its expected running time is how long we should expect the algorithm to take on average, even if it sometimes has fast or slow runs.

Algorithms with a random choice involved in the algorithm are generally judged by their expected running times instead of the worst case. Board game players typically try to make plays that maximize their expected points. Occasionally it's not the right way to think about a problem, especially when hitting particular values could achieve particular effects; in a board game with one move to go, the second-place player really just wants to maximize the chance of beating the first place player, and not pick safe moves with good expected returns that still lose. But, as a first pass, using expectation to decide what to do is the most common approach in all kinds of risky situations.

In essence, the *expectation of a random variable is its average value, where each value is weighted according to the probability that it comes up*. The expectation is also called the *expected value or the mean* of the random variable.

The expected value of a random variable is written as:

$$E[X] = \mu = \sum_{X=x} xP(X = x)$$

In other words, we take the weighted average of the outcomes, where each outcome is multiplied by its probability.

Let's look at an example

If there's a 5% chance of \$100, and a 10% chance of \$50, then the expected reward for the lottery ticket is:

$$E[X] = 0.05*100 + 0.10*50 = 5 + 5 = 10.$$

If we just care about this lottery as a moneymaking opportunity, we should be indifferent to paying for a ticket if the price is \$10, we should walk away if the price is above that, and we should buy as many tickets as we can if the price is below that.

Linearity of Expectation

The expectation of a random variable that is the sum of two random variables, is the sum of their individual expectations:

$$E[X + Y] = E[X] + E[Y]$$

Let's look at an example

If Z is a random variable representing the sum of two dice, then the expected value for Z is calculated as follows:

Let the individual die results be called X and Y . We know, $E[X] = E[Y] = 3.5$

$$E[Z] = E[X] + E[Y] = 3.5 + 3.5 = 7$$

This rule would keep applying as we keep adding variables. So, the expected value of the sum of 10 dice will be $10*3.5 = 35$

Let's practice

- Justin picked up \$300 worth of clothing at Macys. As a part of promotion, he will flip a fair coin at the cash register. If the coin lands on heads, he will get a 40% off on his entire purchase. Otherwise, he will pay regular price. Calculate the expected value of his purchase.

- John just bought a new phone and is considering buying a warranty. The warranty costs \$200 and is worth \$1000 if the phone breaks. He estimates that there is a 10% chance that his phone will break. Should he get the warranty?
- Worldwide oil transport corporation is considering hiring a small oil spill response team to follow one of their ships from Alaska to Russia. The team would cost \$50,000 but would save an estimated \$25,00,000 in the event of an oil spill. The ship is estimated to have a 1% chance of generating an oil spill. Is hiring the team a good decision monetarily?
- You draw one card from a standard deck of playing cards. If you pick a heart, you win \$10. If you pick a face card, which is not a heart, you win \$8. If you pick any other card, you lose \$6. Should you play?
- A detective figures that he has one in nine chances of recovering stolen property. His out-of-pocket expense for the investigation is \$9000. If he is paid his ^{fee} free only if he recovers the stolen property, what should he charge his clients in order to break even?

$$1. P(H) = 1/2,$$

if heads, cost of purchase $\rightarrow 300 \times .6 = 180$

$$P(T) = 1/2$$

if Tails, cost of purchase $\rightarrow 300$

let $E(P) \rightarrow$ Expected value of purchase

$$\therefore E(P) = \frac{1}{2} \times 180 + \frac{1}{2} \times 300 = 90 + 150 = 240$$

2. If the phone breaks,

Benefit from warranty $= \$ (1000 - 200) = \$ 800$

[Total benefit, = (Amount of money the warranty gives - the cost of warranty)

If the phone doesn't break,

Benefit from warranty $= - \$ 200$

$$\therefore E[X] = 0.1 \times 800 - 0.9 \times 200 = - \$ 100$$

Monetarily, he will be losing \$100 in the long run, so he shouldn't buy the warranty.

OR

$$E(X) = -10 \times 1000 + .9 \times 0$$
$$= -100$$

Cost of warranty $\rightarrow 200$

$$\therefore E(X) - \text{cost of warranty}$$
$$= -\$100$$

3. $\rightarrow X \rightarrow$ Random Variable
Calculating Values of X :

$\rightarrow$ If an oil spill happens, total value saved
 $(2500 - 50)$ Thousand dollars.
 $= 2450$ thousand dollars.

$\rightarrow$ If oil spill doesn't happen, value lost
 $\Rightarrow 50$ thousand dollars $\downarrow$
 -50 thousand

→ Create probability distribution table:

X	Value	Probability	Value · Prob.
Oil spill	2450	0.01	24.5
No Oil spill	-50	0.99	-49.5

→ Calculate $E(X)$

$$E(X) = (24.5 - 49.5) \text{ thousand dollars} \\ = -25 \text{ thousand dollars.}$$

∴ Hiring the team will cause an expected loss of 25 thousand dollars in the long run.

4.
$$X = \begin{cases} 10 & \text{if } H \\ 8 & \text{if } F \neq H \\ -6 & \text{if Any other} \end{cases}$$

$X \rightarrow$ Random variable.

3 possible values : $+\$10$, $+\$8$, $-\$6$

Calculating probabilities

$$P(X = \$10) = P(\text{Heart}) = \frac{13}{52}$$

$$P(X = \$8) = P(\text{Face card} \neq \text{Heart}) = \frac{3 \times 4 - 3}{52} = \frac{9}{52}$$

$$P(X = -\$6) = P(\text{remaining cards})$$

$$= \left(1 - \frac{13 + 9}{52} \right)$$

$$= 1 - \frac{22}{52} = \frac{30}{52}$$

Calculating $E(X)$

* you can make a probability distribution table here if you want

$$E(X) = P(X=10) \times 10 + P(X=8) \times 8 + P(X=-6) \times (-6)$$

$$= 10 \times \frac{13}{52} + 8 \times \frac{9}{52} - 6 \times \frac{30}{52}$$

$$= \frac{22}{52} \approx \$0.42$$

∴ There is an expected gain of \$0.42. Therefore, it is to the player's advantage to play the game.

5. Let X be a random variable

→ Here we are interested in the value of X such that Total $E(X) = \$9000$ (only then he breaks even)

Let the fee he charges be $\$x$

$$\therefore E(X) = \frac{1}{9}x + \frac{8}{9} \times 0$$

$$9000 = \frac{x}{9}$$

$$\Rightarrow x = \$81,000$$

OR

Total profit he will make if he finds the stolen goods, given that he charges $\$x$
 $= \$ (x - 9000)$

Breaking even here will mean $E(X) = 0$
(i.e. Expected value of net profit is $\$0$)

$$\therefore E(X) = \frac{1}{9}(x - 9000) - \frac{8}{9} \times 9000$$

$$0 = x - 9000 - 72000$$

$$0 = x - 81000$$

$$\therefore x = \$81000$$